Progress Review Presentation

CM 4900 - Research Project in AI

ADVERSARIAL GEOMETRIC DISTILLATION BASED MITIGATING HALLUCINATIONS IN 3D MODEL GENERATION

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05.05.2026

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Presentation Outline

1. Introduction
2. Objectives
3. Literature Review
4. Problem Definition (Research Gap)
5. Technology Adopted (Extended)
6. Novel Approach
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Introduction

Technological Area : Generative Artificial Intelligence for Text-to-3D Object Generation

Key Achievements :

Built a working prototype for multi-view detection, geometric analysis, and mesh refinement

Challenges : Janus artifacts
Floating parts
Topology errors
Loss of spatial detail in text-based feedback

Technological Research Gap :

Existing text-to-3D methods cannot precisely detect and correct 3D geometry hallucinations in a closed loop

Intended Extension :

A geometry-aware AGD refinement layer that maps detected errors to 3D vertices and selectively corrects the mesh

Research Objectives

- Critical review of the literature on geometric hallucinations in generative 3D models, analyzing how the evolution of these models led to specific topological errors and structural inconsistencies

- In-depth study of adversarial neural networks and self-distillation technologies used for automated artifact detection and mesh refinement

- Design and develop an Adversarial Geometric Distillation system that integrates an auxiliary AI critic to guide the correction of 3D geometric anomalies

- Evaluate the proposed solution using quantitative geometric metrics like surface smoothness, manifold quality and qualitative performance comparisons against baseline models

Literature Review

Chronological Highlights

1. NeRF: Enabled neural 3D scene representation and view synthesis [1]
2. DreamFusion: Introduced SDS for text to 3D generation [2]
3. Magic3D: Improved quality with coarse-to-fine high-resolution generation [3]
4. Fantasia3D: Separated geometry and appearance for better 3D control [4]
5. Perp-Neg / ProlificDreamer: Reduced Janus and mode collapse issues, but still geometry blind [6]
6. Hallo3D: Added multimodal hallucination detection, but still relied on text-based feedback [7]

Literature Review

Challenges Identified:

Study & Year	Key Concept	Primary Limitation(s)
DreamFusion [2] (2022)	SDS-based text-to-3D generation	Geometry-blind optimization; causes Janus and view inconsistency
Magic3D [3] (2023)	Coarse-to-fine high-resolution 3D generation	Better detail, but still inherits 2D-prior structural errors
Fantasia3D [4] (2023)	Disentangled geometry and appearance	Improves quality, but lacks direct topology-aware correction
Perp-Neg [6] (2023)	Score-space / negative prompt debiasing	Reduces Janus only; does not directly fix 3D geometry
Entropic Score Distillation [5] (2024)	Mode-collapse reduction through entropy regularization	Improves diversity, but still no spatially grounded 3D correction
Hallo3D [7] (2024)	Multimodal hallucination detection with prompt feedback	Semantic bottleneck; spatial detail is lost in language-based correction

Technology to Extend: Existing text-to-3D generation will be extended with Adversarial Geometric Distillation (AGD) to bridge the identified research gap

Problem Definition

Current text-to-3D generation frameworks remain fundamentally geometry-blind, as they depend on 2D diffusion priors and semantic feedback loops that cannot convert hallucination detections into precise 3D vertex-level corrections, resulting in persistent structural errors such as Janus artifacts, disconnected geometry, and topological inconsistency

Technology Extended

Technological Area : Generative Artificial Intelligence for Text-to-3D Object Generation with Adversarial Geometric Learning and Geometry-Aware Mesh Refinement.

Fundamentals & Strengths:

- Score Distillation Sampling (SDS)-based Text-to-3D: Excellent at converting text-guided 2D diffusion priors into 3D generation, enabling strong zero-shot synthesis and high visual quality.
- Multi-View Hallucination Detection: Allows the generated 3D object to be inspected from multiple viewpoints, improving the detection of inconsistencies such as Janus artifacts and duplicated structures.
- Spectral and Topological Mesh Analysis: Provides mathematically grounded measures such as Laplacian variance, Fiedler value, connected components, and genus for evaluating structural quality.
- Graph-Based Refinement Signals: Supports localized anomaly scoring at the vertex level, making geometric correction more targeted and controllable.

Technology Extended

Limitations of Current Technology:

- SDS relies on 2D diffusion priors and lacks explicit 3D geometric constraints.
- Existing multimodal methods often pass corrections through language, causing a semantic bottleneck and loss of spatial precision.
- Most current approaches detect hallucinations at the view level or global level, not at the exact vertex or mesh-region level.
- Refinement is often diffuse or generic, so topological defects and disconnected geometry can remain unresolved.

How it is Extended:

- We extract the correction logic out of text-only feedback and move it into a separate geometric grounding and refinement engine that dynamically converts detected hallucinations into vertex-level geometric weights for selective mesh correction.

Novel Approach

Adversarial Geometric Distillation

Hypothesis :

Structural hallucinations in text-to-3D generation can be reduced by Adversarial Geometric Distillation, which combines multi-view hallucination detection, adversarial geometric criticism, and spatially grounded mesh refinement.

Inspiration :

Inspired by Hallo3D, spectral graph theory, algebraic topology, adversarial learning, and self-distillation.

Input :

- Text prompt
- Generated 3D mesh
- Multi-view renders
- Refinement parameters

Novel Approach

Adversarial Geometric Distillation

Output:

- Refined 3D mesh with fewer Janus artifacts
- Floating components
- Topological defects

Technology:

- Extends text-to-3D generation with an adversarial geometric critic
- Spectral-topological analysis
- Multi-view inspection
- Geometry-aware refinement

Novel Approach

Adversarial Geometric Distillation

Process :

1. **Multi-View Renderer**

   24–100 RGB+Depth views via Fibonacci or grid camera placement

2. **Hallucination Detector**

   5-signal ensemble: CLIP, Depth, Normal, Silhouette, Edges

3. **Geometric Critic (AGD)**

   Spectral + VLM analysis → anomaly masks + scores

5. **Distillation Loss**

   $L\_total = \Sigma w_i \cdot L_i$ Added to L_SDS for differentiable signal

6. **ground errors to vertices**

   Iterative vertex optimization guided by geometric loss

Novel Approach

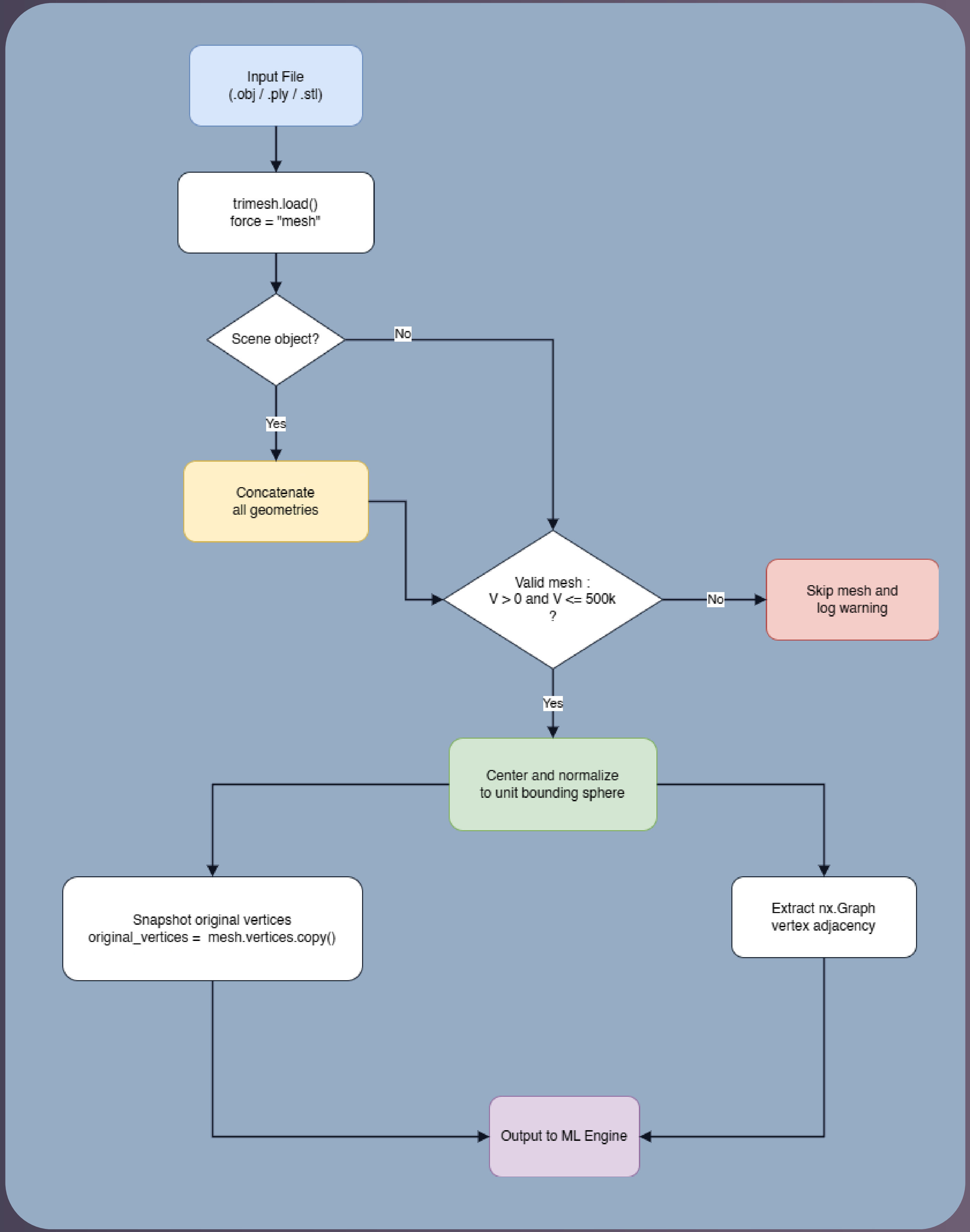
Adversarial Geometric Distillation

Users:

- 3D GenAI researchers
- VR/AR developers
- Game asset creators
- Academic evaluators

Features:

- Closed-loop correction
- Semantic bottleneck removal
- Vertex-level grounding
- Modular design
- Selective refinement



Design

Top-Level Architecture:

A closed-loop architecture separating the analysis layer from the refinement layer

Logical Positioning:

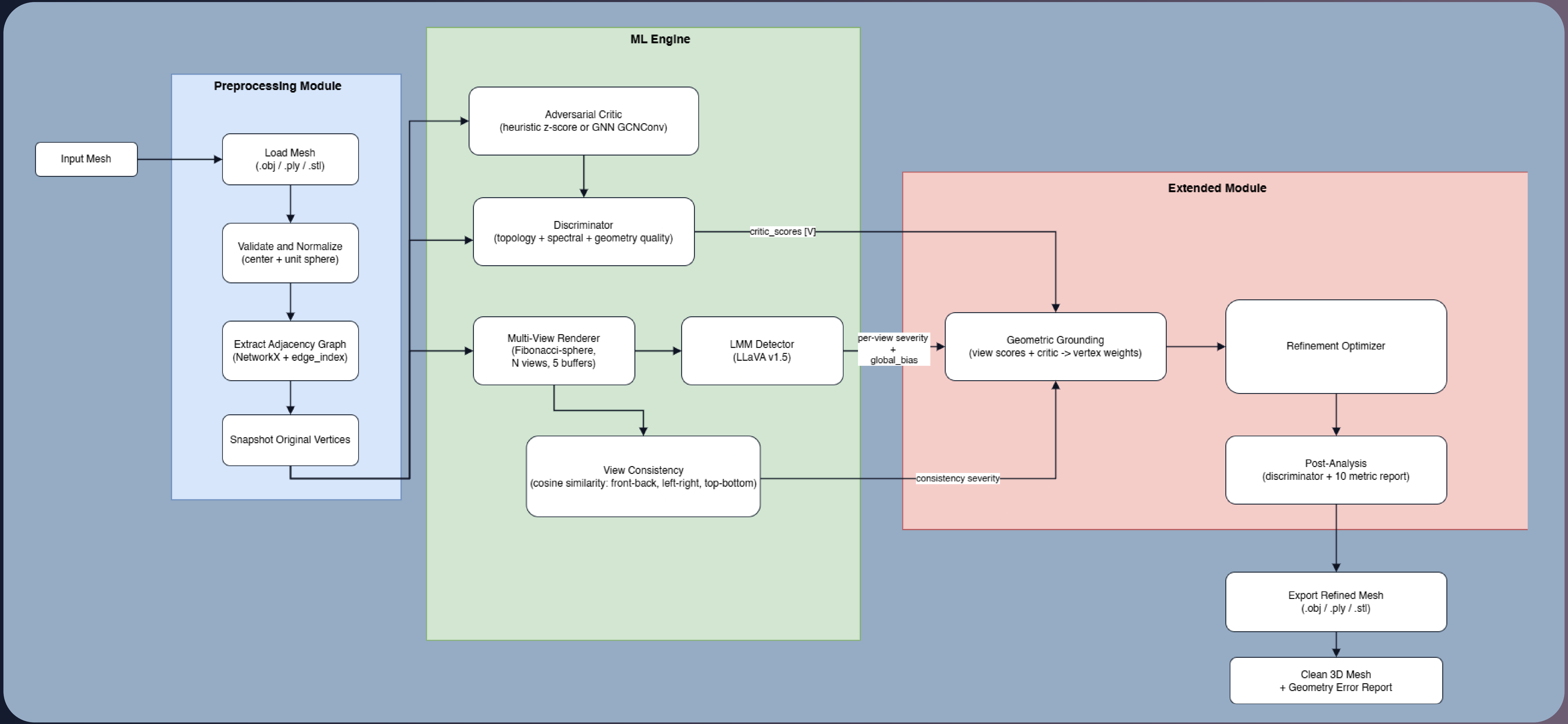
AGD operates as a post-generation geometric correction layer between text-to-3D synthesis and final mesh output

Component Roles & Justification:

- Base Generator: Produces the initial hallucination-prone 3D mesh
- Detection Controller: Manages multi-view inspection and anomaly evaluation
- Adversarial Critic and Grounding Module: Identifies problematic geometry and maps errors to precise mesh regions
- Refinement Module: Applies selective geometry-aware updates to improve structural consistency

Design

Top-Level System Architecture Diagram



Current Implementation

Overall:

Developed as a modular Python-based AGD pipeline for closed-loop hallucination detection, geometric grounding, and mesh refinement

OS / Platforms:

Windows/Linux; geometry analysis runs on CPU, while NVIDIA RTX GPU (CUDA) is used for LLaVA inference and GNN-based extensions

SW / HW:

Python 3.10+, trimesh, numpy, scipy, networkx, torch, torch_geometric, transformers, Pillow, scikit-image.

Algorithms:

Fibonacci-sphere multi-view rendering, spectral/topological mesh analysis, heuristic/GNN adversarial critic, projection-based geometric grounding, and weighted Laplacian + self-distillation refinement

Current Implementation

Codes:

- agd_pipeline.py (main pipeline)
- discriminator.py (topology/spectral scoring)
- critic.py (vertex anomaly scoring)
- renderer.py (multi-view rendering)
- grounding.py (view-to-vertex mapping)
- optimizer.py (mesh refinement)
- lmm_detector.py (LLaVA inspection)

Datasets:

- Curated 3d_samples/ mesh dataset with clean and hallucinated 3D models
- Plus Generated multi-view renders for detection and evaluation

```
AGD Pipeline — 1 meshes │ 10 iters │ lr=0.15

janus.obj

  Loaded: 20,008 verts, 40,000 faces
  BEFORE
    score=0.575  variance=1.16190  fiedler=-0.00000
    components=4  genus=-3.5  euler_char=9.0
INFO: Rendering dense views (n=6, method=grid) …
INFO: Rendering 8 views via 'grid' schedule.
INFO: Rendered 8 views → 40 image files.
  AFTER
    score=0.575  variance=1.16190  fiedler=-0.00000
    components=4  genus=-3.5  euler_char=9.0
  Δ score=+0.000→  Δ variance=+0.00000
INFO: Rendering 8 views via 'grid' schedule.
INFO: Rendered 8 views → 40 image files.
  Computing geometry error metrics …
  ── Geometry Error Metrics (before vs. after) ──
  Chamfer Distance    (↓ better): 0.042993  [fwd 0.021364, bwd 0.021630]
  Hausdorff Distance  (↓ better): 0.072044  [HD95 0.042010]
  Normal Consistency  (↑ better): 0.9223  [|cos| 0.9349]
  Angular Normal Err  (↓ better): mean=15.00°  p95=50.54°  max=158.68°
  Surface Roughness   (↓ better): RMS=0.007096
  Curvature W1        (↓ better): 0.002050
    curvature before=0.009546  after=0.007371
  ── Multi-view scan (8 views) ──
  Mean Silhouette IoU (↑ better): 0.9912
  Mean Depth RMSE     (↓ better): 10.4483
  Mean SSIM           (↑ better): 0.9486
  Mean Edge IoU       (↑ better): 0.6302
  Scan aggregate (worst-case views):
    silhouette_iou_min: 0.9878
    depth_rmse_max: 15.2095
    ssim_min: 0.9437
    edge_iou_min: 0.5483
  Saved → janus_agd.obj  [62.3s]
```

FINAL SUMMARY

Mesh	Score B→A	CD	HD95	NCS	Ang°	Rough	SSIM
janus.obj	0.575→0.575	0.04374	0.04198	0.919	15.28	0.00705	0.9498

Experiment Conducted

Overall: Conducted a preliminary CLIP-based multi-view experiment to test whether hallucinations in 3D meshes can be detected without model training

Why CLIP: CLIP was used because it can measure semantic consistency across different rendered views, making it useful for spotting Janus artifacts and view-level structural drift

Method: Rendered 24 views of a mesh and combined CLIP semantic deviation with geometric signals such as depth discontinuity, silhouette asymmetry, and normal inconsistency

View	Az°	El°	CLIP Dev	Depth Disc	Silh Asym	Composite
0	0	-20	0.039	0.277	0.035	0.089
1	45	-20	0.061	0.233	0.043	0.087
2	90	-20	0.081	0.192	0.120	0.095
3	135	-20	0.060	0.174	0.132	0.085
4	180	-20	0.069	0.295	0.035	0.103
5	225	-20	0.058	0.204	0.043	0.078
6	270	-20	0.062	0.201	0.120	0.090
7	315	-20	0.052	0.219	0.132	0.093
8	0	0	0.049	0.300	0.057	0.101
9	45	0	0.053	0.227	0.062	0.085
10	90	0	0.054	0.242	0.120	0.098
11	135	0	0.054	0.153	0.146	0.079
12	180	0	0.048	0.254	0.053	0.089
13	225	0	0.040	0.166	0.050	0.064
14	270	0	0.051	0.193	0.104	0.082
15	315	0	0.038	0.191	0.060	0.070
16	0	30	0.055	0.240	0.103	0.095
17	45	30	0.061	0.239	0.085	0.094
18	90	30	0.050	0.299	0.051	0.100
19	135	30	0.030	0.196	0.009	0.062
20	180	30	0.061	0.198	0.044	0.078
21	225	30	0.051	0.163	0.077	0.070
22	270	30	0.048	0.263	0.080	0.095
23	315	30	0.053	0.208	0.000	0.071

Experiment Conducted

Key Result: On the Stanford Dragon mesh, the system produced a low global hallucination score of 0.0856, with no major Janus or floater issues detected

Impact on Research: This experiment became the baseline and motivation for the AGD extension, where view-level detection is upgraded into vertex-level geometric grounding and refinement

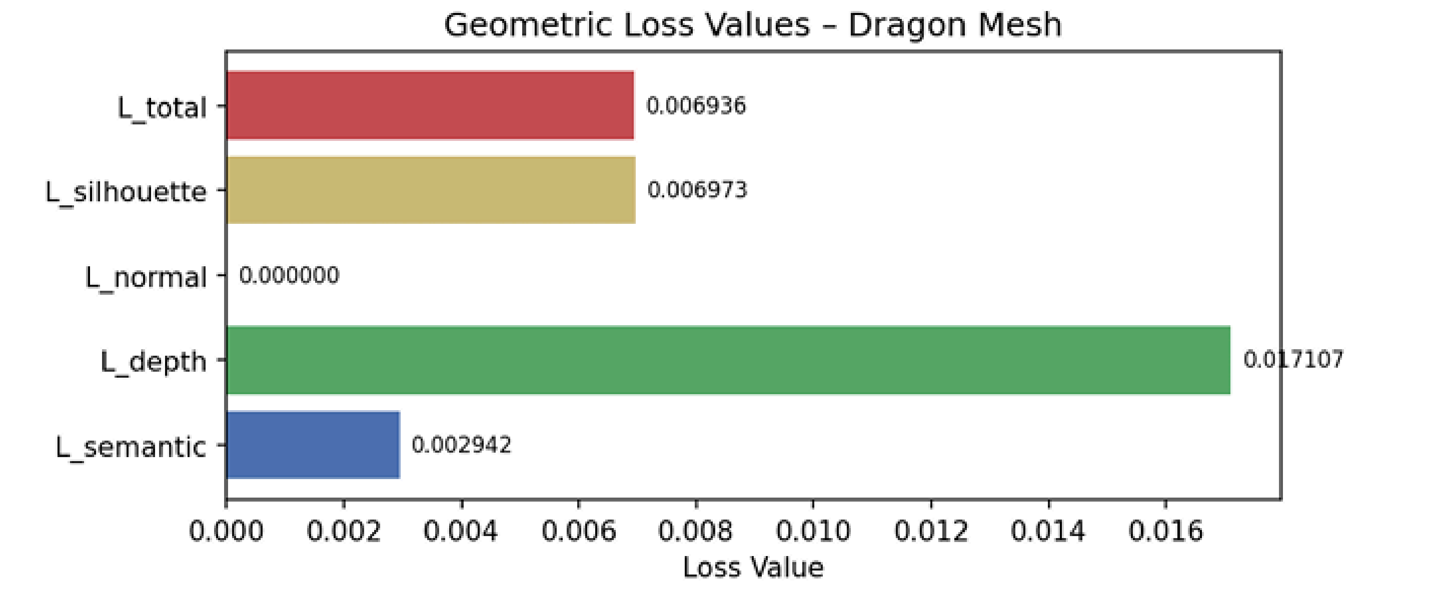


Figure 7 – Computed geometric loss values. L_depth dominates.

References

[1] G. Metzer, E. Richardson, O. Patashnik, R. Giryes, and D. Cohen-Or, “Latent-NeRF for Shape-Guided Generation of 3D Shapes and Textures,” Nov. 14, 2022, arXiv: arXiv:2211.07600. doi: 10.48550/arXiv.2211.07600.

[2] B. Poole, A. Jain, J. T. Barron, and B. Mildenhall, “DREAMFUSION: TEXT-TO-3D USING 2D DIFFUSION”.

[3] Z. Wang et al., “ProlificDreamer: High-Fidelity and Diverse Text-to-3D Generation with Variational Score Distillation,” Nov. 22, 2023, arXiv: arXiv:2305.16213. doi: 10.48550/arXiv.2305.16213.

[4] R. Chen, Y. Chen, N. Jiao, and K. Jia, “Fantasia3D: Disentangling Geometry and Appearance for High-quality Text-to-3D Content Creation,” Sep. 27, 2023, arXiv: arXiv:2303.13873. doi: 10.48550/arXiv.2303.13873.

[5] C.-H. Lin et al., “Magic3D: High-Resolution Text-to-3D Content Creation,” Mar. 25, 2023, arXiv: arXiv:2211.10440. doi: 10.48550/arXiv.2211.10440.

[6] J. Cui et al., “Hallo3: Highly Dynamic and Realistic Portrait Image Animation with Video Diffusion Transformer”.

[7] M. Armandpour, A. Sadeghian, H. Zheng, A. Sadeghian, and M. Zhou, “Re-imagine the Negative Prompt Algorithm: Transform 2D Diffusion into 3D, alleviate Janus problem and Beyond,” Apr. 26, 2023, arXiv: arXiv:2304.04968. doi: 10.48550/arXiv.2304.04968.

[8] P. Wang et al., “Taming Mode Collapse in Score Distillation for Text-to-3D Generation,” in 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA: IEEE, Jun. 2024, pp. 9037–9047. doi: 10.1109/CVPR52733.2024.00863.

[9] T. Wu et al., “Recent advances in 3D Gaussian splatting,” Comp. Visual. Med., vol. 10, no. 4, pp. 613–642, Aug. 2024, doi: 10.1007/s41095-024-0436-y.

[10] H. Fan, H. Su, and L. Guibas, “A Point Set Generation Network for 3D Object Reconstruction from a Single Image,” in 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI: IEEE, Jul. 2017, pp. 2463–2471. doi: 10.1109/CVPR.2017.264.

[11] K. Schwarz, A. Sauer, M. Niemeyer, Y. Liao, and A. Geiger, “VoxGRAF: Fast 3D-Aware Image Synthesis with Sparse Voxel Grids”.

[12] A. Jain, B. Mildenhall, J. T. Barron, P. Abbeel, and B. Poole, “Zero-Shot Text-Guided Object Generation with Dream Fields,” in 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, LA, USA: IEEE, Jun. 2022, pp. 857–866. doi: 10.1109/CVPR52688.2022.00094.

[13] A. Nichol, H. Jun, P. Dhariwal, P. Mishkin, and M. Chen, “Point-E: A System for Generating 3D Point Clouds from Complex Prompts,” Dec. 16, 2022, arXiv: arXiv:2212.08751. doi: 10.48550/arXiv.2212.08751.

[14] H. Jun and A. Nichol, “Shap-E: Generating Conditional 3D Implicit Functions,” May 03, 2023, arXiv: arXiv:2305.02463. doi: 10.48550/arXiv.2305.02463.

[15] Y. Siddiqui et al., “MeshGPT: Generating Triangle Meshes with Decoder-Only Transformers,” in 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA: IEEE, Jun. 2024, pp. 19615–19625. doi: 10.1109/CVPR52733.2024.01855.

[16] J. Yang, K. Fogarty, F. Zhong, and C. Oztireli, “SYM3D: Learning Symmetric Triplanes for Better 3D-Awareness of GANs”.

[17] C. Tsalicoglou, F. Manhardt, A. Tonioni, M. Niemeyer, and F. Tombari, “TextMesh: Generation of Realistic 3D Meshes From Text Prompts,” in 2024 International Conference on 3D Vision (3DV), Davos, Switzerland: IEEE, Mar. 2024, pp. 1554–1563. doi: 10.1109/3DV62453.2024.00154.

Thank You